1.16 Graph Sketching: Quartic Graphs

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Quartics

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If you understand the principle of sketching polynomials in general, then sketching quartics shouldn't feel like anything new.
Recall that if the x^4 term is positive, the 'tails' both go upwards, otherwise downwards.

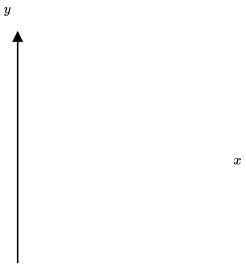
Sketch the curve with equation
$y = x(x+1)(x-2)(x-3)$

Shape:
Roots:
y-intercept:



Sketch the curve with equation
$y = (x-2)^2(x+1)(3-x)$

Shape:
Roots:
y-intercept:



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Quartics

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Sketch the curve with equation
$y = (x+1)(x-1)^3$

-1 root only appears once so line crosses at $x = -1$
+1 root triple repeated so point of inflection at $x = 1$



Sketch the curve with equation
$y = (x-2)^4$

2 is a quadruple repeated root! Because the line effectively crosses the axis 4 times all at -2, it ends up in the opposite direction, and hence looks like a 'touch' point.



1.16 Graph Sketching: Quartic Graphs

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Test Your Understanding

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Sketch the curve with equation
$y = x^2 (x + 1) (x - 1)$

y

x

Sketch the curve with equation
$y = -(x + 1)(x - 3)^3$

y

x

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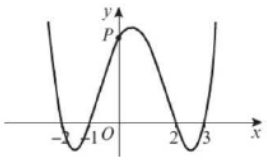
Exercise 4B – Pure Y1 Pages 65-66

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3 The graph of $y = x^4 + bx^3 + cx^2 + dx + e$ is shown opposite, where b, c, d and e are real constants.

a Find the coordinates of point P. **(2 marks)**

b Find the values of b, c, d and e. **(3 marks)**



4 Sketch the graph of $y = (x + 5)(x - 4)(x^2 + 5x + 14)$.

1.16 Graph Sketching: Quartic Graphs

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Exercise 4B Extension

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Extension

1 *[STEP I 2012 Q2a]*
a. Sketch $y = x^4 - 6x^2 + 9$
b. For what values of b does the equation $y = x^4 - 6x^2 + b$ have the following number of distinct roots (i) 0, (ii) 1, (iii) 2, (iv) 3, (v) 4.

a)

b) By changing b, we shift the graph up and down. Then we can see that:

i) 0 roots:
ii) 1 root:
iii) 2 roots:
iv) 3 roots:
v) 4 roots: